

Bayesian Inference

Likelihood: $p(d|\theta) = \frac{1}{\sqrt{(2\pi)^m |C|}} e^{-\frac{1}{2}(d-h(\theta)|d-h(\theta))}$

- probability of observing data d given model parameters θ .

→ Really we want

$p(\theta|d)$: prob. of these model parameters given the data.

posterior distribution

Conditional Probability

$p(A|B) \Rightarrow p(A)$ if B is true

$$p(A|B) \cdot p(B) = p(B|A) p(A)$$

$$p(A|B) = \frac{p(B|A) p(A)}{p(B)}$$

Bayes' Theorem

Rolling two dice

$$d_1 = 1 \dots 6$$

$$d_2 = 1 \dots 6$$

$$s = d_1 + d_2$$

$$p(d_1) = \frac{1}{6} \text{ for all options}$$

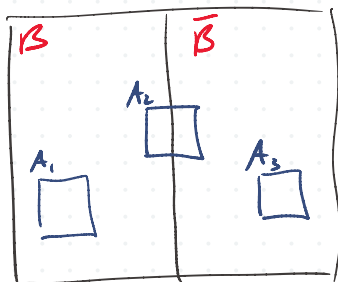
$$p(d_2) \quad " \quad " \quad " \quad "$$

$$p(s=7) = \sum_{d_1+d_2=7} p(d_1)p(d_2) = \frac{6}{36}$$

$$p(s=7 | d_1=1) = \frac{1}{6}$$

$$p(d_1=1 | s=7) = \frac{1}{6}$$

1	6
2	5
3	4
4	3
5	2
6	1



$$P(B) = 0.5$$

$$P(A_1) = P(A_2) = P(A_3) = 0.04$$

$$P(B|A_1) = 1$$

$$P(B|A_2) = 0.5$$

$$P(B|A_3) = 0$$

$$P(A_1|B) = P(B|A_1) \cdot P(A_1) / P(B)$$

$$= 1 \cdot 0.04 / 0.5 = 0.08$$

$$P(A_2|B) = 0.5 \cdot 0.04 / 0.5 = 0.04$$

$$P(A_3|B) = 0 \cdot 0.04 / 0.5 = 0$$

In Bayesian inference,

$$p(\theta | d) = \frac{p(d|\theta) p(\theta)}{p(d)}$$

posterior ← likelihood → prior

Evidence

$$p(\theta | d, M) = \frac{p(d|\theta, M) p(\theta|M)}{p(d|M)}$$

All these probabilities depend on our choice of model.

Can compare the evidence of two models to know which is a better explanation of the data.

$p(\theta|M)$: the prior. This does not depend on the data, what we know before the experiment.

$p(d|\theta, M)$: the likelihood. How well the parameters fit the data

$p(d|M)$: the evidence. How well the model fits the data (for any parameters).

$$p(d|M) = \int d\theta \, p(d|\theta, M) p(\theta|M)$$

$p(\theta|d, M)$: the posterior. What we know after the experiment.

Bayesian Learning

Start with prior

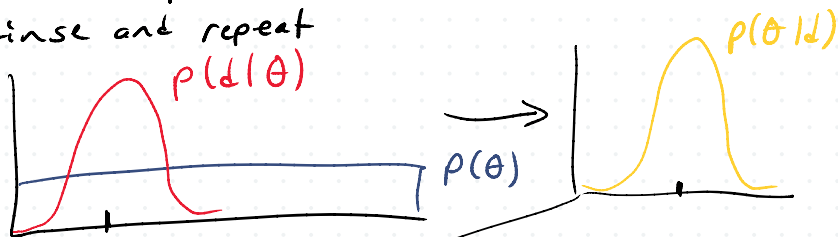
Add data

Get posterior

Today's posterior is tomorrow's prior

Rinse and repeat

$$\begin{array}{l} p(\theta) \\ + d \\ \hline p(\theta|d) \end{array}$$



More data $\Rightarrow$ converge to true parameters

$$p(\theta|d_1) = \frac{p(d_1|\theta) p(\theta)}{p(d_1)}$$

$$p(\theta|d_1, d_2) = \frac{p(d_1, d_2|\theta) p(\theta)}{p(d_1, d_2)}$$

Probability chain rule: $p(A, B) = p(A|B) \cdot p(B)$

$$p(d_1, d_2|\theta) = \underbrace{p(d_1|\theta, d_2)}_{p(d_1|\theta)} \cdot p(d_2|\theta)$$

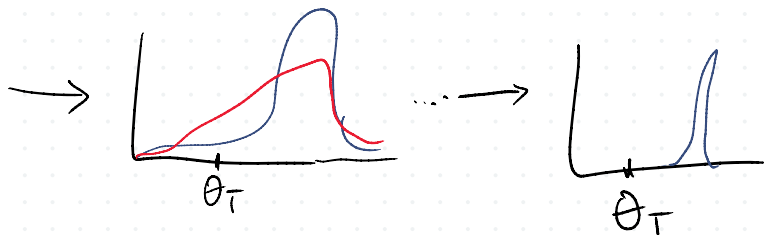
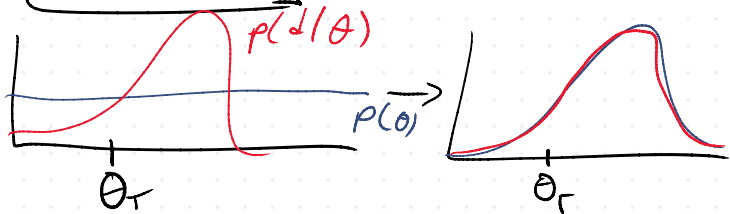
d_1 and d_2 are independent!
 $= p(d_1|\theta) \cdot p(d_2|\theta)$

$$p(d_1, d_2) = p(d_1) \cdot p(d_2)$$

$$p(\theta | d_1, d_2) = \frac{p(d_2|\theta) \cdot \boxed{p(\theta|d_1)}}{p(d_2)}$$

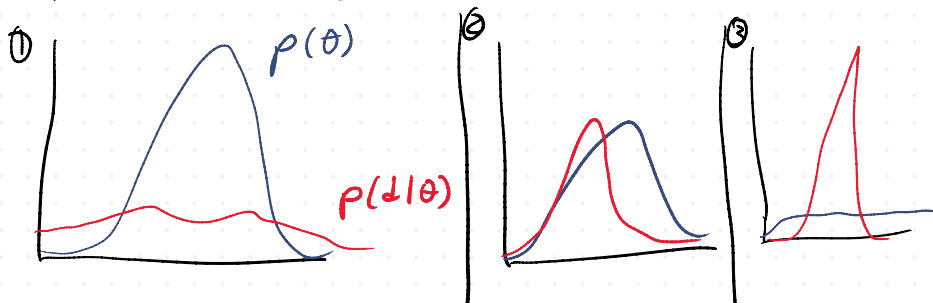
Why not use d to update the prior?

Don't do it!



You only get one chance to learn from the data. Prior should be independent of the data.

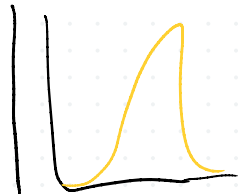
Three interesting regimes:





$p(\theta|d)$

prior-dominated!
Data are uninformative
⇒ More data!



"Hard" Data + prior
equally informative.
Needs careful thought



"Likelihood dominated"
Answer is mostly
indep. of prior.

Could have same posterior, but very different interpretations!

How much have we learned from the data?
Measure the "distance" between $p(\theta)$ and $p(\theta|d)$

Kull back-Leiber (?) Divergence (KL-divergence).

$$D_{KL}(P, Q)$$

↑
prior

↑
posterior

$$D_{KL}(P, Q) \equiv \sum_x P(x) \log\left(\frac{P(x)}{Q(x)}\right)$$

① $p(\theta|d) \sim p(\theta)$: $D_{KL} \sim 0$

② UGH

③ $p(\theta|d) > p(\theta) \rightarrow D_{KL} \gg 0$.

Posterior

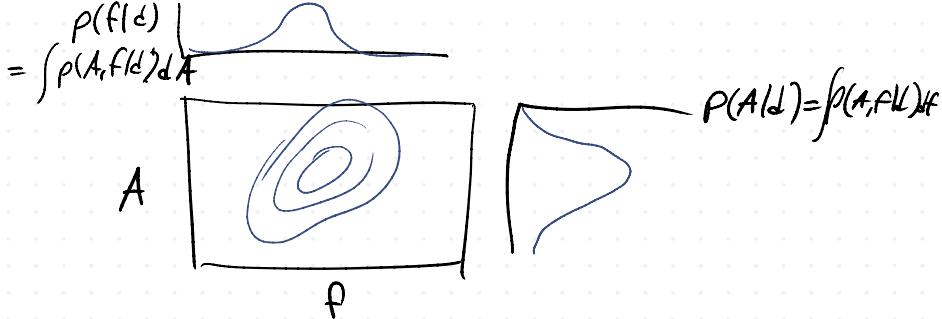
$$M: h(\theta) = A \sin(2\pi f t + \phi_0)$$

$$\theta = \{A, f, \phi_0\}$$

⇒ $p(A, f, \phi_0 | d)$ is our posterior.

3D probability distribution!

$$p(A, f | d) = \int p(A, f, \phi_0 | d) d\phi_0 \quad \text{"Marginalizing"}$$



Marginalized distributions are easier to look at, but we lose information!